

Human Digestive System

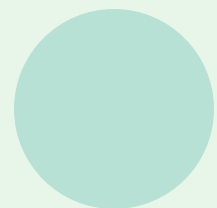


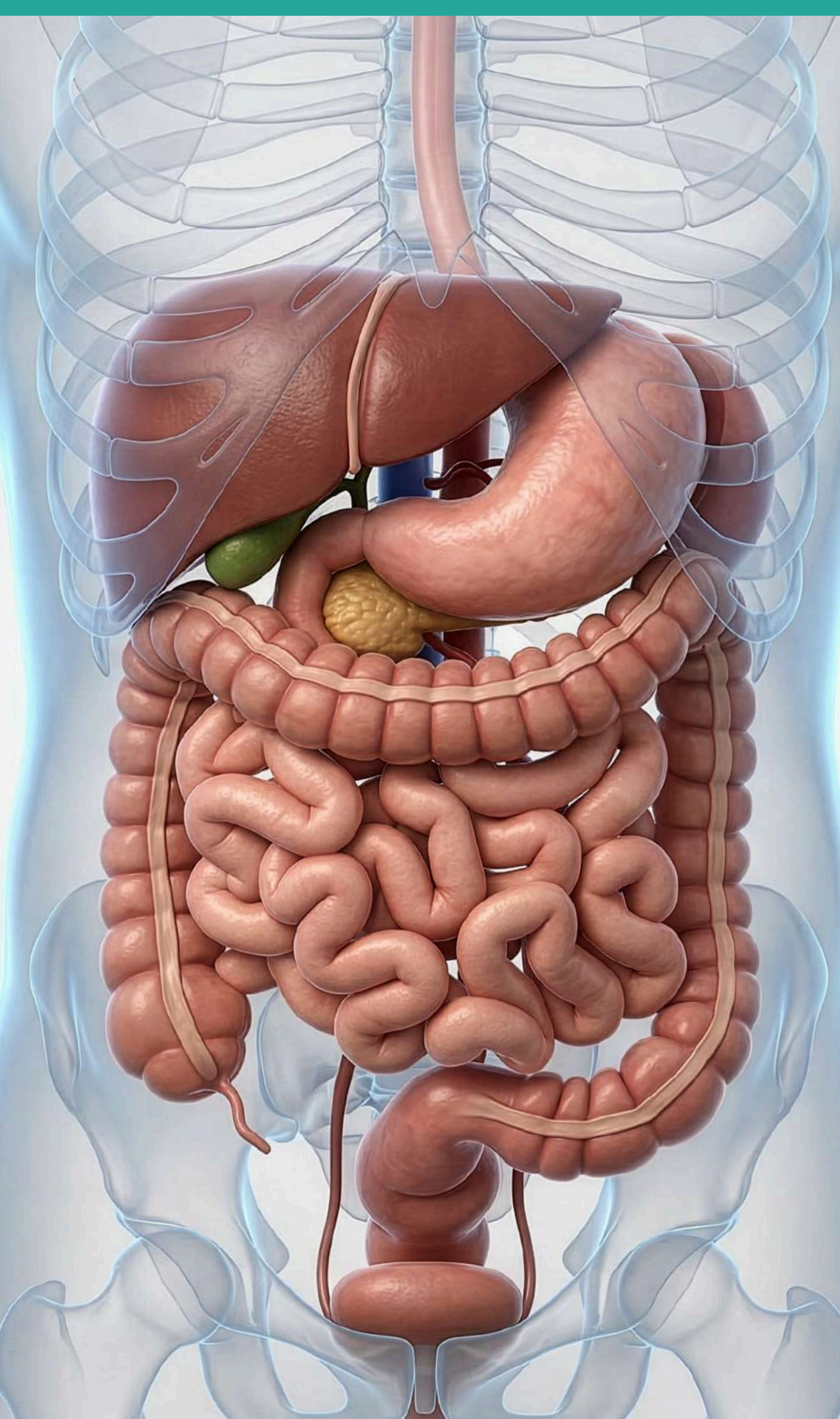
What Happens During Digestion?

THE BASICS OF DIGESTION

Digestion is how our body breaks down food to get energy and nutrients.

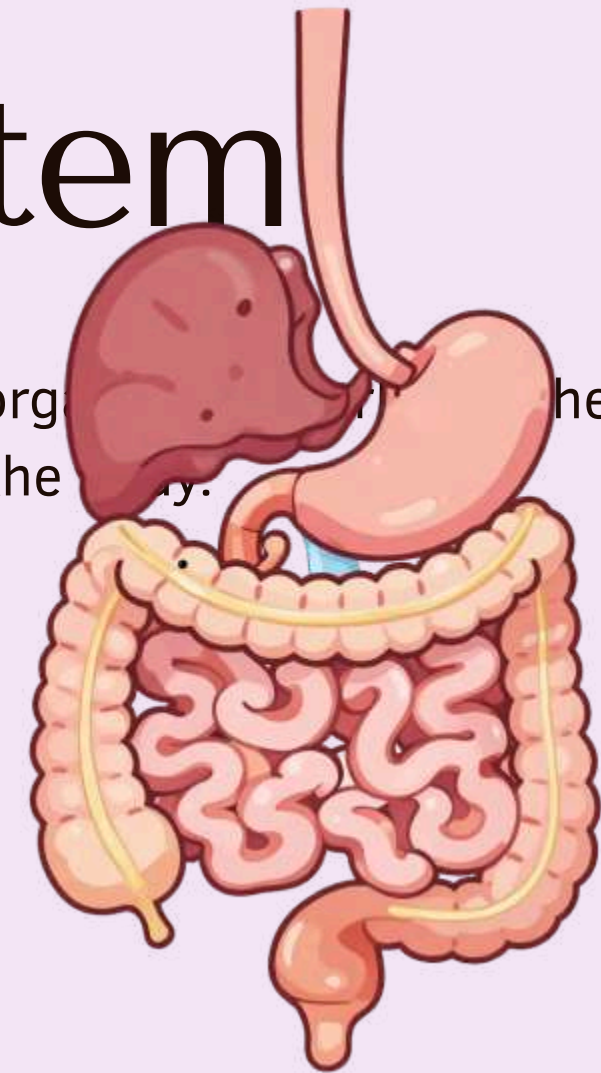
- It starts in the mouth.
- Enzymes help break down food.
- Nutrients are absorbed in the small intestine.
- Our body uses nutrients for growth and energy.





Discover the Main Organs of the Digestive System

The digestive system is a **complex network** of organs that work together to break down food and absorb essential nutrients into the body.



What Are Enzymes?

TINY HELPERS IN DIGESTION

Enzymes are **tiny helpers** that break down food into smaller parts.

- They act like scissors, cutting food into tiny pieces.
- Without enzymes, our bodies would struggle to get energy.
- Enzymes work at different stages of digestion.

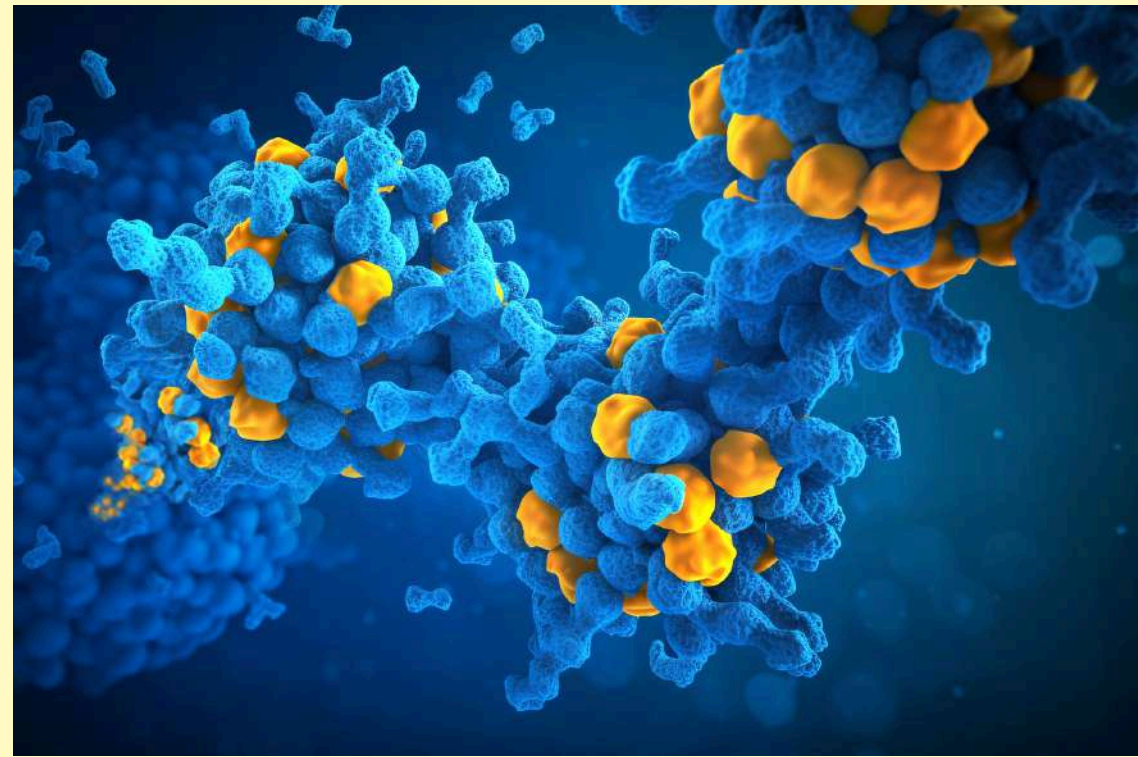


Amylase: The Enzyme That Cuts Starch



SALIVA

Amylase is found in our saliva.



STARCH

It breaks down starch into simple sugars.



DIGESTION

You start digesting food while chewing!

Pepsin: The Protein Breaker Enzyme

PROTEIN DIGESTION IN ACTION

Pepsin is an **essential enzyme** that breaks down proteins into smaller amino acids. It works perfectly in the stomach's acidic environment.

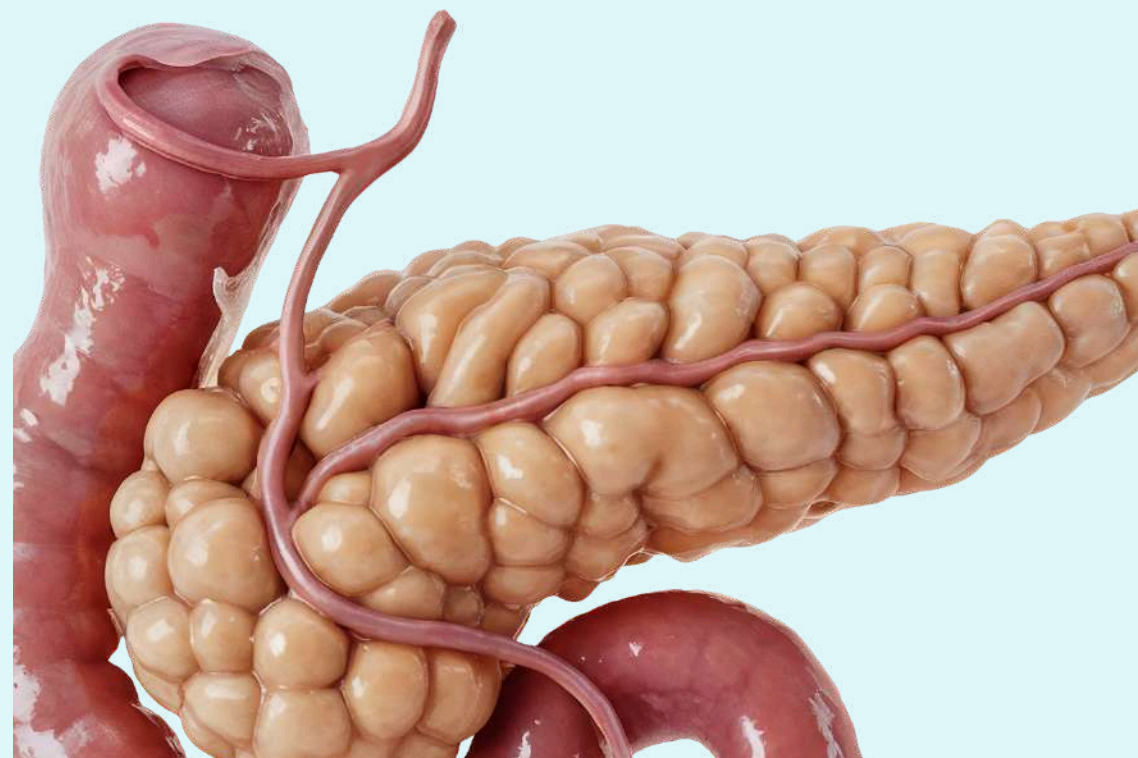


Understanding Lipase — The Fat Fighter



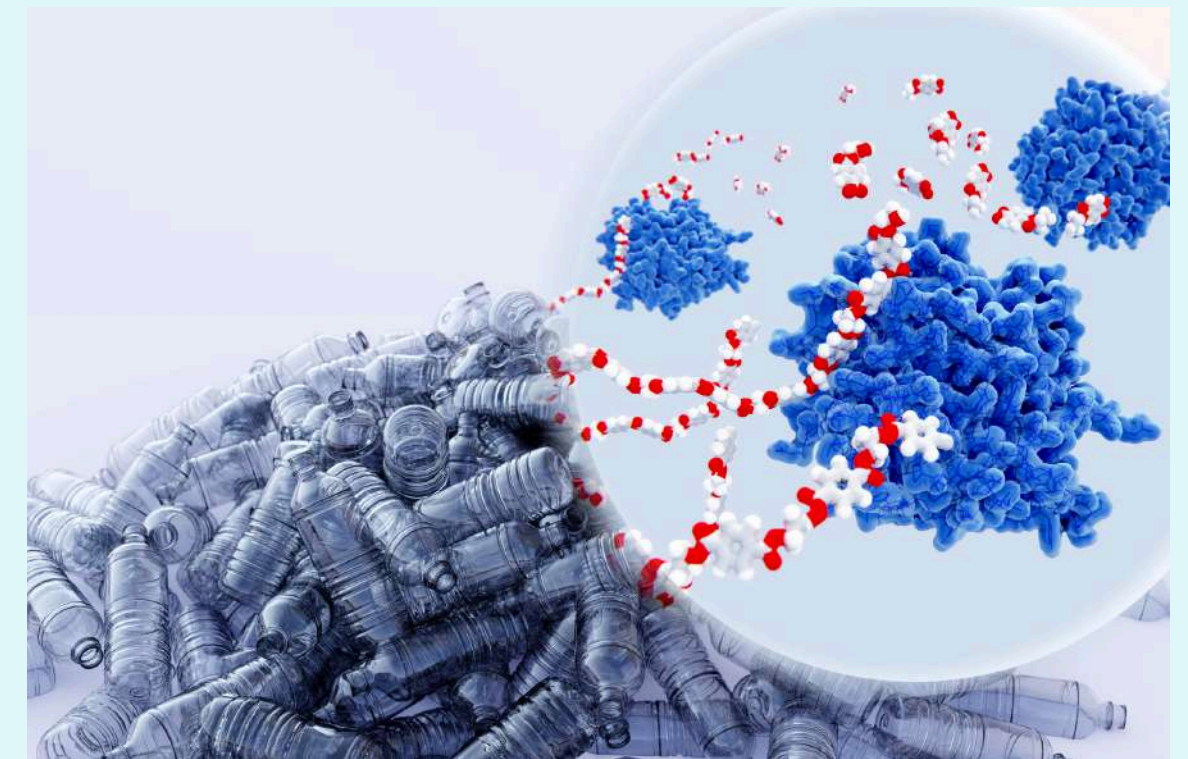
SOURCE

Lipase is produced in the pancreas.



FUNCTION

It breaks down fats into fatty acids efficiently.



IMPORTANCE

Essential for nutrient absorption and energy production.

How Enzymes Break Down Food

THE ROLE OF ENZYMES

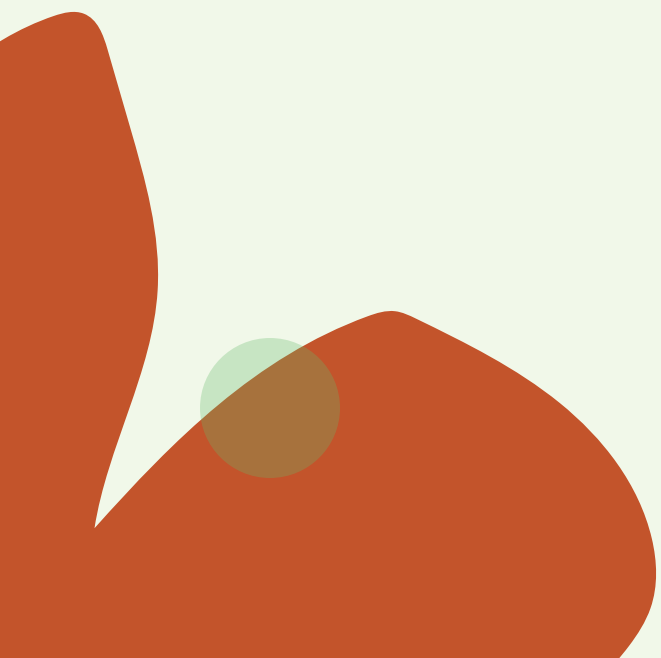
Enzymes are essential **catalysts** that speed up the breaking down of food, making digestion quicker and more efficient for our bodies.

BREAKING DOWN MOLECULES

By cutting large food molecules into smaller, absorbable pieces, enzymes help our body maximize nutrient intake from the food we eat.

SPECIFIC FUNCTIONS

Each enzyme has a specific job, ensuring that starch, protein, and fat are broken down into their simplest forms for absorption.



The Food's Journey

MOUTH

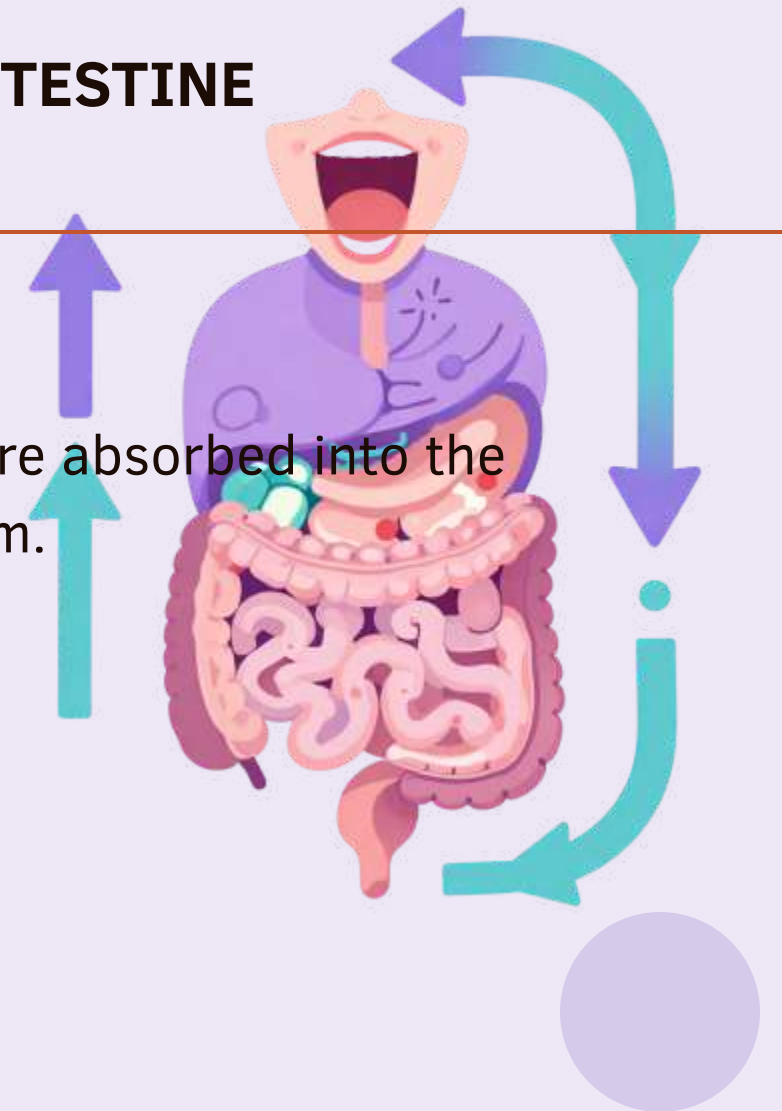
Food enters the mouth and begins digestion.

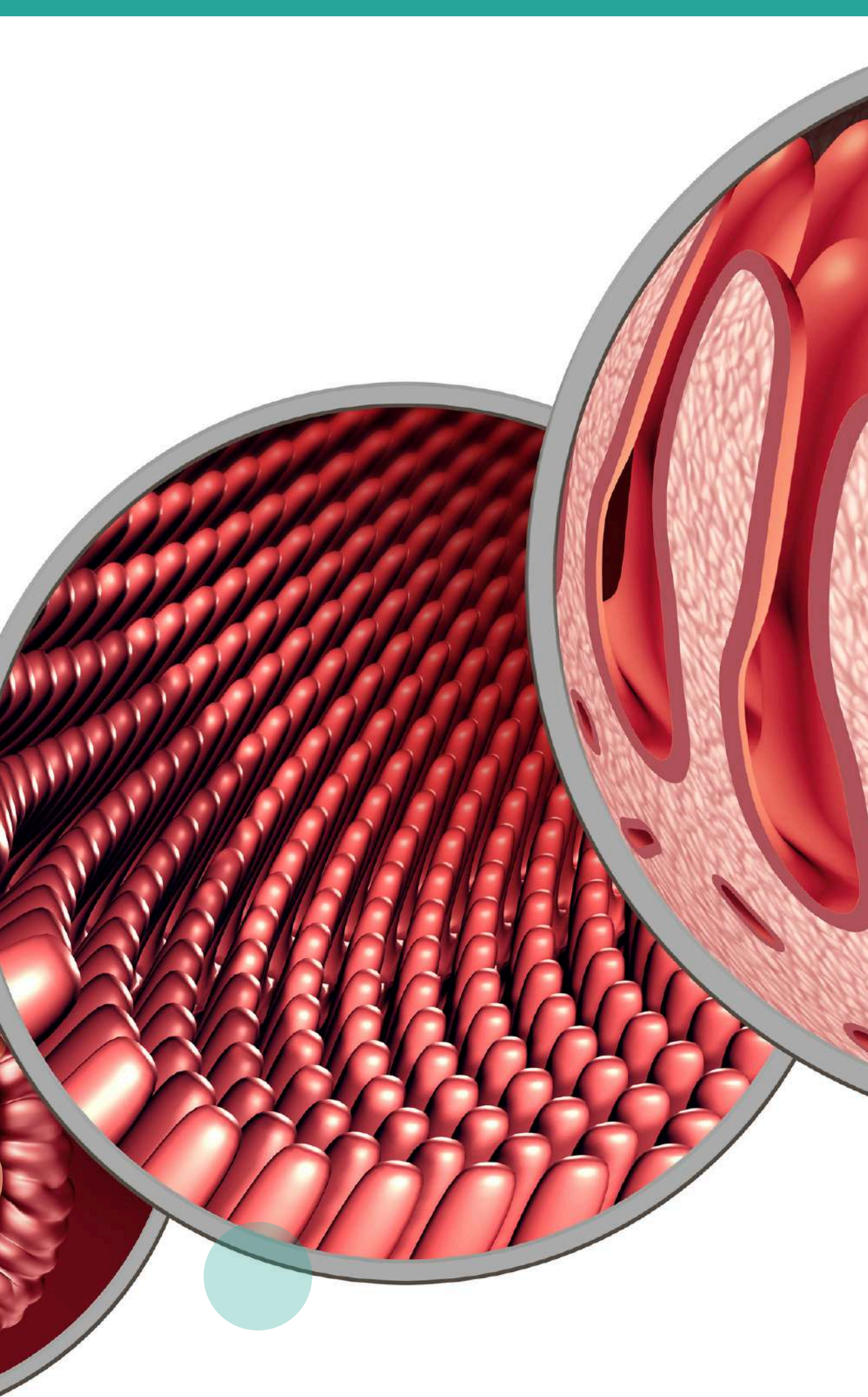
STOMACH

Food is mixed with acids and enzymes here.

SMALL INTESTINE

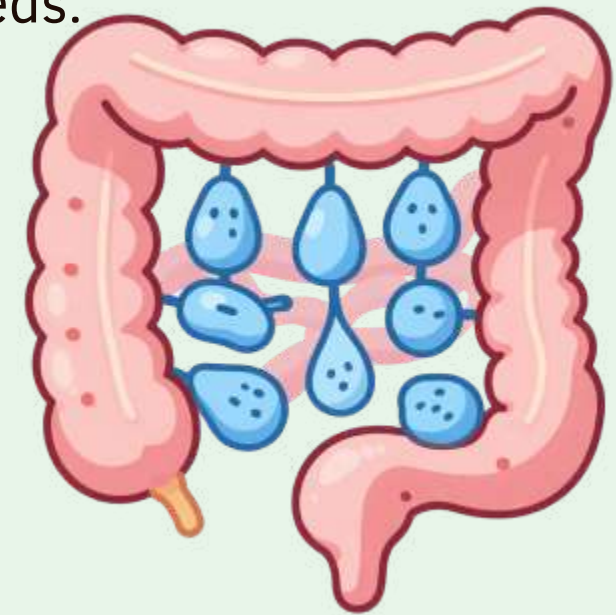
Nutrients are absorbed into the bloodstream.





How Nutrients Are Absorbed in the Small Intestine

The small intestine's walls have tiny **finger-like villi** that absorb nutrients into the blood, ensuring your body gets the energy it needs.





6 meters

LENGTH OF SMALL INTESTINE

Your small intestine is about **6 meters long**, allowing for maximum absorption of nutrients. This extensive length provides ample surface area for tiny food particles to pass through the walls into the bloodstream, ensuring your body receives the essential nutrients it needs to grow and stay healthy.



Recap: Enzymes and Absorption

KEY POINTS TO REMEMBER ABOUT DIGESTION

- Enzymes help break down food into nutrients
- Amylase, Pepsin, and Lipase are important enzymes
- Absorption primarily occurs in the small intestine
- Nutrients enter the bloodstream to fuel your body



TEST YOUR KNOWLEDGE!

- Which enzyme breaks down starch?
- Where does pepsin work?
- What part of the digestive system absorbs nutrients?
- How do enzymes help digestion?
- Share your answers with the class!



GET READY FOR FUN!

Answer the questions from the quiz and discuss with your classmates. Remember to think about what you learned during the presentation. Each answer helps build your understanding of the **amazing digestive system** and its functions. Enjoy sharing your knowledge!



Fun Fact About Your Stomach

A SURPRISING TRUTH!

Did you know that your stomach makes a **new lining every 3 days** to protect itself from its own acids?

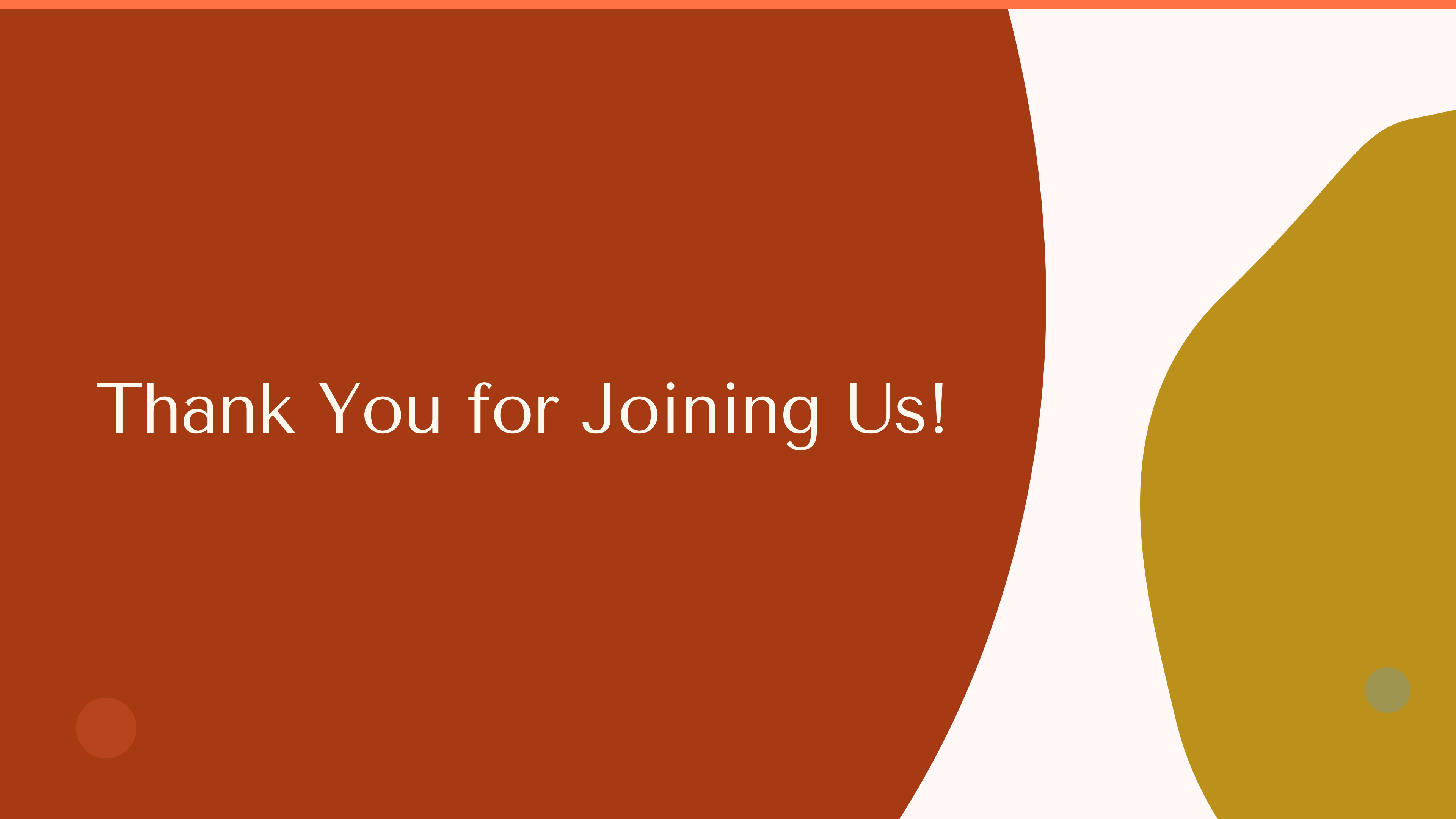


Summary of Key Concepts

IMPORTANT TAKEAWAYS ABOUT DIGESTION

- Digestion turns food into energy
- Enzymes make digestion process easier
- Absorption happens in small intestine
- Nutrients fuel your body's needs
- Keep your digestive system healthy





Thank You for Joining Us!